Logic, Language, and Philosophy

Intensional logic

Meeting 27, December 22, 2025

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Overview

- Recap
- Intensionality
- Relational semantics
- Modal proposition logic
  - Modal languages and their semantics
  - Modal logics
- Other intensional logics
- Quantifiers and domains
- Logic as tool
- Homework

Recap

Extensional semantics

‘What an expression means is what it refers to’

Proposition logic:

- Formulas: truth values
- Connectives: truth functions

Predicate logic:

- Vocabulary: individuals; sets of (n-tuples of) individuals
- Connectives: truth functions
- Quantifiers: truth functional [caveat]

Extensionality theorem

Theorem (Extensionality)

$$\forall x_1 \ldots x_n(\phi \leftrightarrow \psi) \models \chi \leftrightarrow [\psi/\phi]\chi$$

where $x_1 \ldots x_n$ are the variables that occur free in ϕ and ψ

Intensionality

Sources

- Logic
- Concepts
- Natural language

Logic

- Material implication and conditional reasoning
  - paradoxes of material implication $\rightsquigarrow$ strict implication
  - failure of weakening $\rightsquigarrow$ non-monotonic logic, default reasoning
- Negation and disjunction
  - constructivist reasoning $\rightsquigarrow$ abandon double negation, excluded middle

$\Rightarrow$ *Different logics require different semantics*

Concepts

- Time
- Modality
- Deontic concepts
- Epistemic concepts

$\Rightarrow$ *These concepts are about what is actual and non-actual*:

- Time: what will be the case, need not be the case now
- Modality: what is possible need not be actual
- Deontic: what is required need not be true
- Epistemic: what is believed, need not be true

$\Rightarrow$ *Different philosophical perspectives are possible*

- Time: (in)determinism of the future?
- Modality: is what is necessarily possible?
- Deontic: is what required possible?
- Epistemic: positive introspection? negative introspection?

$\Rightarrow$ *Different perspectives can be formalised as different logics*

Natural language

- Functional expressions, predicates about their range of values
  - The temperature is 90°, the temperature is rising / 90° is rising
- Temporal and modal expressions
  - The number of planets is eight, eight is necessarily even / the number of planets is necessarily even
- Deontic and epistemic expressions
  - *x* knows that the King of NL is W-A, The King of NL is the chairman of the State Council / *x* knows that the chairman of the State Council is WA

⇒ *The key function of natural language is to allow us go beyond the actual*

Relational semantics

Relational frames

Definition (Relational frame)

A pair $\langle \mathcal{S}, \mathcal{R} \rangle$, with $\mathcal{S}$ a parameter set and $\mathcal{R}$ a relation on $\mathcal{S}$

- Choice of $\mathcal{S}$: determines *what the language is about*
- Choice of $\mathcal{R}$: determines *the properties* of what the language is about

Examples:

- $\mathcal{S}$ = set of temporal points T, $\mathcal{R}$ = ordering E defined on T
- $\mathcal{S}$ = set of possible worlds W, $\mathcal{R}$ = accessibility relation R defined on W

Relational models

Definition (Relational model)

A triple $\langle \mathcal{S}, \mathcal{R}, \mathcal{V} \rangle$, with $\langle \mathcal{S}, \mathcal{R} \rangle$ a relational frame and V a valuation

Example: language of propositional tense logic

Relational model $M = \langle T, E, V \rangle$.

This parameterises the truth definition:

- $V_{M,t}(p) \in \{0, 1\}$, for all proposition letters p
- $V_{M,t}(\neg\phi) = 1$ iff $V_{M,t}(\phi) = 0$
- $V_{M,t}(\phi \wedge \psi) = 1$ iff $V_{M,t}(\phi) = V_{M,t}(\psi) = 1$

Further needed: temporal expressions that exploit that

- $V_{M,t}(F\phi) = 1$ iff there exists a t': Ett' and $V_{M,t'}(\phi) = 1$
- $V_{M,t}(P\phi) = 1$ iff there exists a t': $Et't$ and $V_{M,t'}(\phi) = 1$

Validity

Definition (Validity)

A formula ϕ is valid iff for all M, t: $V_{M,t}(\phi) = 1$

Example: propositional tense logics

Different specifications of E define different *tense logics*:

- E is linear ordering
- E is partial order
- ...

The ground structure of intensional logics

- A specification of the parameter set: this determines what the logic ‘is about’
- A specification of the structure of the parameter set: this allows us to explore variations of the logic
- A set of expressions that are interpreted using the parameter set and its structure
- A truth definition to specify those interpretations

Modal proposition logic

Modal proposition logic

Modal languages and their semantics

Modal languages

A modal propositional language is a propositional language with the addition of two operators, viz., $\Box$ and $\Diamond$:

- If ϕ is a formula, then so are $\Box\phi, \Diamond\phi$

Note that this allows for formulae in which modal operators are nested, such as $\Box\Box p, \Box\Diamond(p \to q), \neg\Box p \to \Diamond(p \vee \Diamond q)$, etc.

Relational frames

A relational frame for a modal propositional language is a pair $\langle W, R \rangle$ with:

- W a non-empty set
- R a relation on W

W is referred to as the set of possible worlds, R as the accessibility relation between worlds.

Note: we write wRw' instead of Rww'

Semantics

A relational model M for a modal propositional language is a triple $\langle W, R, V \rangle$ such that:

- $W_M \neq \emptyset$
- $R_M \subseteq W \times W$
- $V_M \in \text{FORM} \times W \to \{0, 1\}$ such that:
  1. $V_{M,w}(p) \in \{0, 1\}$, for all proposition letters p
  2. $V_{M,w}(\neg\phi) = 1$ iff $V_{M,w}(\phi) = 0$
  3. $V_{M,w}(\phi \wedge \psi) = 1$ iff $V_{M,w}(\phi) = V_{M,w}(\psi) = 1$
  4. $V_{M,w}(\Box\phi) = 1$ iff $V_{M,w'}(\phi) = 1$ for all w' s.th. wRw'
  5. $V_{M,w}(\Diamond\phi) = 1$ iff $V_{M,w'}(\phi) = 1$ for some w' s.th. wRw'

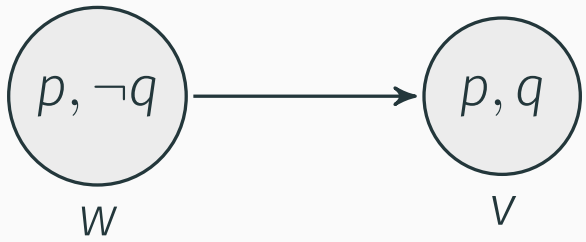
Validity

Definition (Validity)

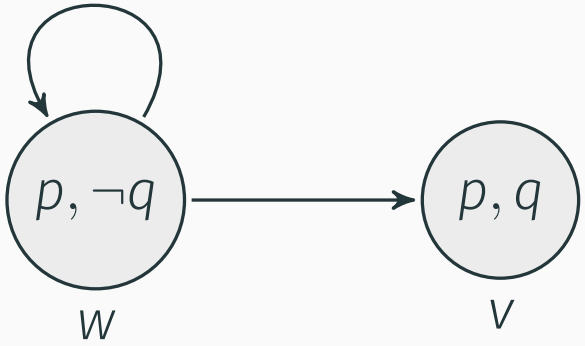
A formula ϕ is valid iff for all M, w: $V_{M,w}(\phi) = 1$

Model checking (1)

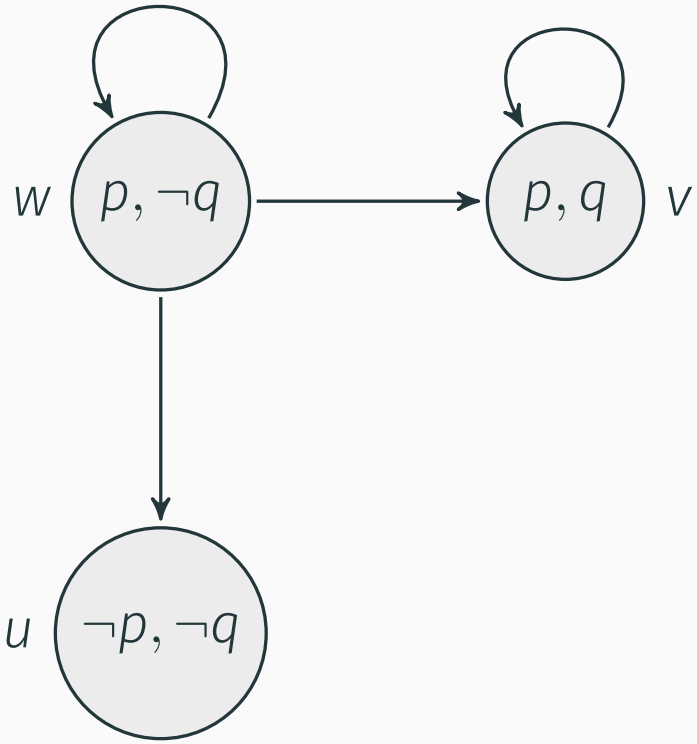
Example 1:



Model checking (2)



Model checking (3)



Limit cases

1. $R = \{\langle w, w\rangle \mid w \in W\}$
2. $R = \emptyset$

Ad 1:

- all $\Box\phi$, $\Diamond\phi$, ϕ become equivalent

Ad 2:

- all $\Box\phi$ become tautologies
- all $\Diamond\phi$ become contradictions

Modal proposition logic

Modal logics

Modal logic K

The following modal principles are valid in all modal models:

- N $\Box\phi$, for all propositional tautologies ϕ
- K $\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$

Modal principles and accessibility

Correspondences between modal principles and properties of the accessibility relation:

- Reflexive ($\forall x Rxx$): $\Box\phi \to \phi$
- Serial ($\forall x \exists y Rxy$): $\Box\phi \to \Diamond\phi$
- Symmetric ($\forall x \forall y (Rxy \to Ryx)$): $\phi \to \Box\Diamond\phi$
- Transitive ($\forall x \forall y \forall z ((Rxy \wedge Ryx) \to Rxz)$): $\Box\phi \to \Box\Box\phi$
- Euclidean ($\forall x \forall y \forall z ((Rxy \wedge Rxz) \to Ryz)$): $\Diamond\phi \to \Box\Diamond\phi$

Some standard modal logics

Name	conditions on R	valid principles
D	serial	$\Box\phi \to \Diamond\phi$
T	reflexive	$\Box\phi \to \phi$
B	reflexive, symmetric	$\Box\phi \to \phi$, $\phi \to \Box\Diamond\phi$
K4	transitive	$\Box\phi \to \Box\Box\phi$
S4	reflexive, transitive	$\Box\phi \to \phi$, $\Box\phi \to \Box\Box\phi$
S5	reflexive, symmetric, transitive	$\Box\phi \to \phi$, $\phi \to \Box\Diamond\phi$, $\Box\phi \to \Box\Box\phi$

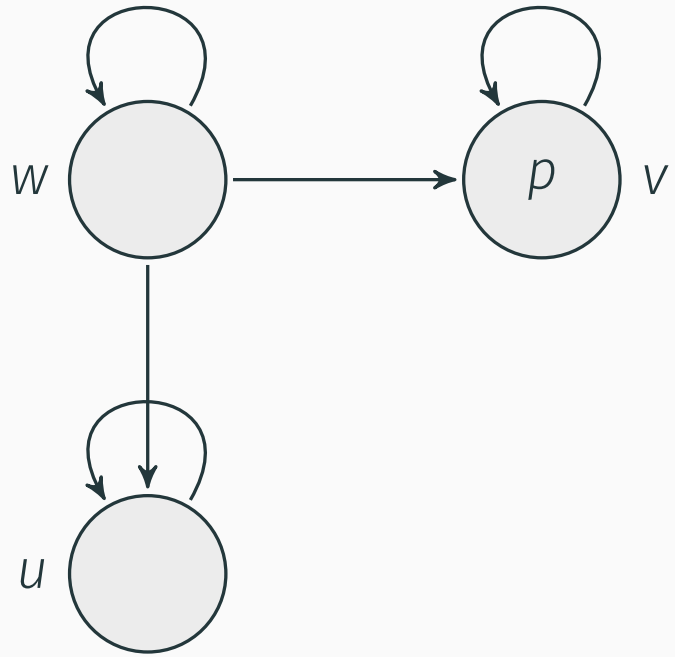
Semantics for intuitionistic logic

An intuitionistic model for a propositional language is a triple $\langle W, R, V \rangle$ such that:

- R is a pre-order of W, i.e., a reflexive and transitive relation on W
- $V \in \text{FORM} \times W \to \{0, 1\}$ that satisfies the following conditions:
  - If $V_w(p) = 1$ and wRw', then $V_{w'}(p) = 1$ (persistence)
  - For no w: $V_w(\bot) = 1$
  - $V_w(\neg\phi) = 1$ iff $V_w(\phi \to \bot) = 1$
  - $V_w(\phi \wedge \psi) = 1$ iff $V_w(\phi) = 1$ and $V_w(\psi) = 1$
  - $V_w(\phi \vee \psi) = 1$ iff $V_w(\phi) = 1$ or $V_w(\psi) = 1$
  - $V_w(\phi \to \psi) = 1$ iff for all w' s.th. wRw' : if $V_{w'}(\phi) = 1$ then $V_{w'}(\psi) = 1$

Semantics for intuitionistic logic (2)

Failure of excluded middle:



$V_w(p \vee \neg p) \neq 1, \neq 0$

Other intensional logics

Interpreting $\Box$ and $\Diamond$

One can assign different interpretations to the $\Box$- and $\Diamond$-operators:

- deontic:
  - $\Box\phi$: It is obligatory that ϕ
  - $\Diamond\phi$: It is permitted that ϕ
- epistemic:
  - $\Box\phi$: It is known that ϕ
  - $\Diamond\phi$: It is consistent with what is known that ϕ

On the epistemic interpretation, the operators and the accessibility relation are usually parameterised to individuals:

- $\Box_x\phi$: x knows that ϕ
- wR_xw': w' is epistemically accessible for x from w

Of principles and plausibility

Not all principles are plausible under all interpretations of the $\Box$- and $\Diamond$-operators.

Examples:

- $\Box\phi \to \phi$:
  - modal: plausible
  - deontic: not plausible
  - epistemic: plausible
- $\Box\phi$, if ϕ is valid:
  - modal: plausible
  - deontic: neither plausible, nor implausible
  - epistemic: not plausible

Quantifiers and domains

Models for first order modal languages

Definition (Frames)

A frame F is a triple $\langle W, R, \{D_w \mid w \in W\}\rangle$, with W, R as usual, and for each $w \in W$ a set of individuals D_w ('the domain of w')

Definition (Model)

A model M is a frame with a parametrised interpretation function I_w such that $I_w(c) \in D_w$, $I_w(P) \subseteq D_w$, and so on.

Modalities *de dicto* and *de re*

- *De dicto*: no occurrence of a modal operator contains free occurrence(s) of variable(s)
- *De re*: at least one occurrence of a modal operator contains at least one free occurrence of a variable

Cf.:

- *De dicto*: $\Box\forall xPx$, $\Box\forall xPx \rightarrow \exists yRyz$
- *De re*: $\forall x\Box Px$, $\Box(\forall xPx \rightarrow \exists yRyz)$

Question: what do quantifiers that bind variables in the scope of a modal operator range over? Just the 'local' individuals? All individuals in all accessible domains?

Example

Suppose:
there are $d, w, w' : wRw', d \in D_w, d \notin D_{w'}$

Then:
$d \notin I_{w'}(P)$ since $I_{w'}(P) \subseteq D_{w'}$

Question:
does this falsify $\forall x \Box Px$ in w?

Increasing domains

'Increasing Domains Assumption':

- For all w, w': if wRw', then $D_w \subseteq D_{w'}$

This forces the existence of what is actual in all other worlds that are taken into consideration.

The Barcan formulae

Relations between world-relative domains and the accessibility relation correspond to validity of certain formulae:

BF $\forall x \Box Px \rightarrow \Box \forall x Px$: this formula is valid iff domains of accessible worlds do not grow, i.e., iff for all w, w': if wRw', then $D_{w'} \subseteq D_w$

CBF $\Box \forall x Px \rightarrow \forall x \Box Px$: this formula is valid iff domains of accessible do not shrink, i.e., iff for all w, w': if wRw', then $D_w \subseteq D_{w'}$

Accepting BF is associated with 'actualism': there are no 'merely possible' objects

One domain

- All worlds have one and the same domain of possible individuals
- Quantifiers range over the entire domain

Quantification over existing individuals via use of ‘existence’ predicate E which applies in a world w to those entities in D that actually exists in w: $I_w(E) \subseteq D$

Logic as tool

Pluralism and what logic is good for

Observations:

- Question of pluralism with regard to intensional logics is moot: it is their plurality that makes them useful
- Different logics explore different alternative conceptions of a certain phenomenon (modality, time, ...)
- That needs three instruments:
  a. A shared and sufficiently expressive language
  b. A shared set of general semantic concepts
  c. Different ways of applying those concepts

Homework

Homework

- Read chapter 24 and make exercises 1(a), 2(1,3), 3(1,2)
- Read chapter 25, Frege's 'On Sense and Reference' and Kripke's 'Identity and Necessity'